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Semiconductor device and method of fabricating the same

Abstract

The present disclosure provides a semiconductor device and a method of fabricating the same, and the semiconductor device includes a substrate, active areas, and an isolation structure. The active areas are parallel and separately disposed with each other in the substrate, and each of the active areas includes an active fin and active ends disposed at two sides of the active fin. The active fin and the active ends include different materials. The isolation structure is disposed in the substrate to surround the active areas. With this arrangement, the extending area of the active areas may be improved, so as to make sure the storage node contacts formed subsequently may directly and stably contact with the active areas.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure generally relates to a semiconductor device and a method of fabricating the same, and more particularly, to a semiconductor device having active areas and shallow trench isolations and a method of fabricating the same.

2. Description of the Prior Art

(2) With the miniaturization of semiconductor devices and the complexity of integrated circuits, the size of elements is continuously shrinking and the structure is constantly changing. Therefore, maintaining the performance of small-sized semiconductor elements is the standard purpose of the present industry. In the semiconductor fabricating process, most of the active areas are defined on the substrate as a bass element, and then, the required elements are further formed on the active areas. Generally, the active areas are plural patterns formed within the substrate through the photolithography and etching processes. However, due to the sized-shrinking requirements, the width of the active areas has been gradually reduced, and the pitch between the active areas has also been gradually reduced thereby, so that, the fabricating process of active areas encounters plenty limitations and challenges that fails to meet the practical product requirements.

SUMMARY OF THE INVENTION

(3) One of the objectives of the present disclosure provides a semiconductor device and a method of fabricating the same, in which the active areas thereof includes active fins and active ends disposed at two sides of each active fin and including different materials. With these arrangements,

the semiconductor device of the present disclosure enables to enlarge the extending area of the active areas, thereby making surface the storage node contacts (SNC) being directly and stably contacted with the active areas. In this way, the structure of the storage node contacts may have improved structural stability, so as to achieve better device performance.

(4) To achieve the purpose described above, one embodiment of the present disclosure provides a semiconductor device including a substrate, active areas, and an isolation structure. The active areas are parallel and separately disposed with each other in the substrate, and each of the active areas includes an active fin and active ends disposed at two sides of the active fin. The active fin and the active ends include different materials. The isolation structure is disposed in the substrate to surround the active areas.

(5) To achieve the purpose described above, one embodiment of the present disclosure provides a method of fabricating a semiconductor device including the following steps. Firstly, a substrate is provided. Then, a plurality of active areas and an isolation structure are formed in the substrate, the isolation structure surrounds the active areas, wherein each of the active areas includes an active fin and active ends disposed at two sides of the active fin, and the active fin and the active ends include different materials.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 to FIG. 7 are schematic diagrams illustrating a fabricating method of a semiconductor device according to a first embodiment in the present disclosure, wherein:

(2) FIG. 1 is a schematic top view of a semiconductor device after forming active area units in the present disclosure;

(3) FIG. 2 is a schematic cross-sectional view taken along a cross-line A-A' in FIG. 1;

(4) FIG. 3 is a schematic top view of a semiconductor device after forming openings in the present disclosure;

(5) FIG. 4 is a schematic cross-sectional view taken along a cross-line A-A' in FIG. 3;

(6) FIG. 5 is a schematic cross-sectional view of a semiconductor device after performing an epitaxial growing process in the present disclosure;

(7) FIG. 6 is a schematic top view of a semiconductor device after forming an insulating layer in the present disclosure; and

(8) FIG. 7 is a schematic cross-sectional view taken along a cross-line A-A' in FIG. 6.

(9) FIG. 8 is a schematic diagram illustrating a semiconductor device performing an epitaxial growing process according to another preferable embodiment in the present disclosure.

(10) FIG. 9 is a schematic diagram illustrating a semiconductor device after performing an epitaxial growing process according to another preferable embodiment in the present disclosure.

(11) FIG. 10 is a schematic diagram illustrating a semiconductor device according to one preferable embodiment in the present disclosure.

(12) FIG. 11 to FIG. 12 are schematic diagrams illustrating a fabricating method of a semiconductor device according to a second embodiment in the present disclosure, wherein:

(13) FIG. 11 is a schematic cross-sectional view of a semiconductor device after forming an insulating layer in the present disclosure; and

(14) FIG. 12 is a schematic cross-sectional view of a semiconductor device after performing an etching process in the present disclosure.

(15) FIG. 13 to FIG. 15 are schematic diagrams illustrating a fabricating method of a

semiconductor device according to a third embodiment in the present disclosure, wherein:

(16) FIG. 13 is a schematic top view of a semiconductor device after forming active area units in the present disclosure;

(17) FIG. 14 is a schematic cross-sectional view taken along a cross-line A-A' in FIG. 13; and

(18) FIG. 15 is a schematic cross-sectional view of a semiconductor device after performing an epitaxial growing process in the present disclosure.

DETAILED DESCRIPTION

(19) To provide a better understanding of the presented disclosure, preferred embodiments will be described in detail. The preferred embodiments of the present disclosure are illustrated in the accompanying drawings with numbered elements. In addition, the technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.

(20) Please refer to FIGS. 1-7, which are schematic diagrams illustrating a fabricating method of a semiconductor device 300 according to the first embodiment in the present disclosure, with FIGS. 1, 3, and 6 respectively illustrating a top view of the semiconductor device 300 during various forming processes, with FIGS. 2, 4, 5, and 7 respectively illustrating a cross-sectional view of the semiconductor device 300 during various forming processes. Firstly, a substrate 100 is provided, the substrate 100 for example includes a silicon substrate, a silicon containing substrate (such as SiC or SiGe), or a silicon-on-insulator (SOI) substrate, and a first insulating layer 120 is disposed in the substrate 100, to define a plurality of active area units 110 in the substrate 100. The active area units 110 are parallel and separately extended along a direction D1, and which are alternately arranged with each other, wherein the direction D1 is for example intersected with and not perpendicular to the y-direction or the x-direction, as shown in FIG. 1. In one embodiment, each of the active area units 110 may have the same length L1 in the direction D1, and adjacent ones of the active area units 110 may have the same gap "g" therebetween.

(21) In one embodiment, the formation of the active area units 110 may but not limited be accomplished by the following patterning process. For example, a mask layer (not shown in the drawings) may be firstly formed on the substrate 100, with the mask layer including a plurality of mask patterns 101 for defining the active area units 110 and with a portion of the substrate 100 being exposed from the mask layer, an etching process is then performed by using the mask layer, to remove the portion of the substrate 100 and to form at least one shallow trench 102, and an insulating material (not shown in the drawings) for example including silicon oxide (SiO₂), silicon nitride (SiN) or silicon oxynitride (SiON) is formed to fill in the shallow trench 102, to form the first insulating layer 120 with a coplanar surface 120a with the top surface 101a of the mask layer, and to define the active area units 110 simultaneously, as shown in FIG. 1 and FIG. 2. In one embodiment, the formation of the active area units 110 may also be accomplished by a self-aligned double patterning (SADP) process or a self-aligned reverse patterning (SARP) process, but not limited thereto.

(22) Please refer to FIGS. 3-4, a mask layer 130 is formed on the substrate 100 while the mask patterns 101 are still remained on the substrate. The mask layer 130 includes a plurality of openings 131, the openings 131 are respectively in alignment with the gap "g" disposed between adjacent ones of the active area units 110, to expose the first insulating layer 120 and end portions 111 of the active area units 110 underneath. The end portions 111 for example refer to the side edges of each active area unit 110 extending in the y-direction in a top view as shown in FIG. 3. In the present embodiment, the end portions 111 at least includes the side edges of each active area unit 110 extending in the y-direction, and a portion of at least one side edge extending in the direction D1 and adjacent to the side edges of each active area unit 110 extending in the y-direction, as shown in FIG. 3. Then, an etching process is performed through the mask layer 130, to remove the mask patterns 101 remained on the end portions 111 of the active area units 110 thereby exposing partial top surface 110a of the active area units 110, and to partially remove the exposed portions of the

first insulating layer **120**, with the portions of the first insulating layer **120** obtaining a top surface **120b** lower than the top surface **110a** of the active area units **110** after the etching process, as shown in FIG. 4. Accordingly, the top surfaces **110a** and partial sidewalls **110b** of the end portions **111** of each active area unit **110** may be exposed after performing the etching process.

(23) Please refer to FIG. 5, a selectively epitaxial growing process is performed while the mask layer **130** and the mask patterns **101** are still remained on the substrate **100**, to form active ends **113**. Please also refer to FIGS. 5-6, in the present embodiment, the active ends **113** are formed on the side edges extending in the y-direction of each active area unit **110**, and the portion of the at least one side edge extending in the direction D1 and adjacent to the side edges extending in the y-direction, thereby presenting in an L-shape as being viewed from a top view as shown in FIG. 6. In addition, the active ends **113** are formed on the exposed top surface **110a** and partial sidewalls **110b** of the end portions **111** of each active area unit **110**, thereby also presenting in an L-shape as being viewed from a cross-sectional view as shown in FIG. 5, but not limited thereto. On the other hand, the rest portions of each active area unit **110** without epitaxial growing form an active fin **115**, so that, the active ends **113** and the active fin **115** together form an active area **150**, and a plurality of the active areas **150** is formed in the substrate **100**. It is noted that, the active ends **113** for example include an epitaxial material which is different from that of the substrate **100**. For example, while the substrate **100** includes a silicon substrate, the active ends **113** may include silicon germanium (SiGe), but is not limited thereto. Then, the mask layer **130** and the mask patterns **101** are completely removed.

(24) Please refer to FIGS. 6-7, a second insulating layer **140** is formed on the portion of the first insulating layer **120**, so that, the top surface **140a** of the second insulating layer **140** may be coplanar with the top surface (namely, the top surface **110a**) of the active fins **115**, as shown in FIG. 7. Accordingly, the first insulating layer **120** and the second insulating layer **140** may together form an isolation structure **160** in the substrate **100**, to surround the active areas **150**, wherein the second insulating layer **140** is disposed between adjacent ones of the active areas **150**, thereby being surrounded by the first insulating layer **120**, as shown in FIG. 6. With these arrangements, the isolation structure **160** may further isolate the adjacent ones of the active areas **150**, to achieve better isolation. Also, the top surface of the active ends **113** may be slightly higher than the top surface of the active fins **115** and the top surface **140a** of the second insulating layer **140**, to obtain an obvious height difference h1 therebetween, as shown in FIG. 7. On the other hand, the active ends **113** may also include a different roughness from that of the substrate **100** (namely, active fins **115**). For example, while the substrate **100** includes a silicon substrate to have a relative smaller roughness, the active ends **113** (for example including silicon germanium) may include a relative greater roughness. However, the practical roughness of the substrate **100** and the active ends **113** may be diverse by material difference, and which is not limited to above mentioned.

(25) Through the aforementioned processes, the semiconductor device **300** of the first embodiment in the preset disclosure is completed. The semiconductor device **300** for example includes a plurality of the active areas **150** and the isolation structure **160** surrounding the active areas **150**. The active area **150** includes the active fins **115** and the active ends **113** disposed at two sides of the active fin **115**, with the active ends **113** having a different material from that of the active fin **115**, so as to obtain an enlarge length L2 in the direction D1. Accordingly, while forming other elements on the active areas **150** in the subsequent processes, the enlarge length and the epitaxial material achieved by the active ends **113** may provide better electrically connection and more stable contact therebetween.

(26) However, people in the art should fully realize that the semiconductor device and the fabricating method thereof are not be limited to the aforementioned embodiment and may include other examples or may be achieved through other strategies to meet practical product requirements. For example, in one embodiment, while performing the selectively epitaxial growing process, active ends **117** having a relative greater thickness may be formed by increasing the growing area

of the epitaxial materials. Then, the top surface of the active ends **117** may be obviously higher than the top surface (namely, the top surface **110a**) of the active fins **115** and the top surface **140a** of the second insulating layer **140**, thereby obtaining a relative greater height difference **h2**, as shown in FIG. **8**. With these arrangements, the active areas **150a** (including the active ends **117** and the active fins **115**) may include an enlarged length **L3** in the direction **D1**, and also, the height difference **h2** between the active fins **115** and the active ends **117** may also be further increased to enlarge the contact area between the active areas **150a** and the storage node contacts, thereby obtaining a more reliable structure. Otherwise, in another embodiment, while performing the etching process through the mask layer **130**, the mask patterns **101** remained on the end portions **111** of the active units **110** may be removed, and the exposed first insulating layer **120** may also be partially removed, to only expose the top surface of the end portions **111** of the active area units **110**. Namely, after the etching process, a top surface **120c** of a portion of the first insulating layer **120** may be coplanar with the top surface **110a** of the active area units **110**, without exposing the sidewalls **110b** of the end portions **111** of the active area units **110**, as shown in FIG. **9**. Through these arrangements, active ends **213** may be formed only on the top surface **110a** of the end portions **111** of the active area units **110** in the selectively epitaxial growing process performed subsequently, thereby obtaining a linear shape in a cross-sectional view, as shown in FIG. **9**. Thus, through the arrangements in aforementioned two embodiments may also improve the extending area of the active areas, to ensure the directly and stably connection between the active areas and the storage node contacts.

(27) People in the art should fully realize that the semiconductor device and the fabricating method thereof may include other examples or may be achieved through other strategies to meet practical product requirements. The following description will detail the different embodiments of the semiconductor device and the fabricating method thereof. To simplify the description, the following description will detail the dissimilarities among the different embodiments and the identical features will not be redundantly described. In order to compare the differences between the embodiments easily, the identical components in each of the following embodiments are marked with identical symbols.

(28) Please refer to FIG. **10**, which illustrate a schematic diagram of a semiconductor device **500** according to a preferable embodiment in the present disclosure. The structure of the semiconductor device **500** in the present embodiment is substantially the same as the semiconductor device **300** in the aforementioned first embodiment, including the substrate **100**, the active areas **150** (including the active fins **115** and the active ends **113**), and the isolation structure **160** (including the first insulating layer **120** and the second insulating layer **140**), and which will not be redundantly described hereinafter. The difference between the present embodiment and the aforementioned first embodiment is in that the semiconductor device **500** additionally includes a plurality of first wires **540** formed in the substrate **100**, and a plurality of second wires **560** and a plurality of plugs **590** formed on the substrate **100**.

(29) Precisely speaking, the first wires **540** for example are parallel extended along the y-direction, to intersect with the active areas **150** and to pass through the first insulating layer **120** and the second insulating layer **140** at the same time. In one embodiment, a plurality of trenches (not shown in the drawings) which are parallel and separately extended along the y-direction are firstly formed in the substrate **100**. Then, an interface dielectric layer **541** entirely covering surfaces of each of the trenches, a gate dielectric layer **543** covering bottom surfaces of each of the trenches, a gate electrode layer **545** filling up the bottom of each of the trenches, and a mask layer **547** filling up the top of each of the trenches, are sequentially formed in the trenches. Accordingly, the topmost surface of the mask layer **547** may be coplanar with the top surface (the top surface **110a**) of the active fins **115**, as shown in FIG. **10**, so that, each of the first wires **540** embedded in the substrate **100** may therefore function like a buried word line (WL) of the semiconductor device **500**, with each word line (namely the first wire **540**) across the active fins **115** of active areas **150** for

receiving or transmitting voltage signals from each memory cell (not shown in the drawings). Although the extending directions of the trenches or the first wires **540** are not precisely illustrated in the drawings, people well skilled in the arts should fully realize the first wires **540** extended in the y-direction may intersect with the active areas **150** to pass through the isolation structure **160** (including the first insulating layer **120** and the second insulating layer **140**) from a top view.

(30) On the other hand, the second wires **560** are for example parallel extended along the x-direction, to intersect with the active areas **150**, and being perpendicular with the first wires **540** in a projection direction (not shown in the drawings). The second wires **560** and the plugs **590** are alternately arranged with each other on a dielectric layer **580** disposed on the substrate **100**, and the adjacent ones of the plugs **590** and the second wires **560** are isolated from each other by a spacer **570** disposed therebetween, as shown in FIG. **10**. In one embodiment, the spacer **570** for example includes a first spacer **570** (including a material like silicon nitride), a second spacer **573** (including a material like silicon oxide), and a third spacer **575** (for example including a material like silicon nitride), but is not limited thereto. Each of the second wires **560** for example includes a semiconductor layer (for example including polysilicon) **561**, a barrier layer **563** (for example including titanium and/or titanium nitride), a conductive layer **565** (for example including a low-resistant metal like tungsten, aluminum, or copper), and a capping layer **567** (for example including silicon oxide, silicon nitride, or silicon oxynitride), but is not limited thereto. It is noted that, a plurality of contacts (bit line contacts, BLC) **560a** may be formed under the second wires **560**, to further extend into the active fins **115** in the substrate **100**, between two adjacent ones of the first wires **540**, and also between two adjacent ones of active ends **113**. In the present embodiment, the contacts **560a** and the semiconductor layer **561** of the second wires **560** are monolithic, to directly contact the active fins **115**, but not limited thereto. The plugs **590** are directly in contact with the active fins **115**, the active ends **113**, and a portion of the capping layer **547** of the first wires **540**, thereby obtaining a more stably structure of the storage node contact (SNC). It is also noted that, while the thickness of the active ends **113** is obviously greater than that of the active fins **115**, the bottom surface of the plugs **590** may also obtain the corresponding height difference $h1$. Then, the structure of the plugs **590** may become more reliable.

(31) Through the aforementioned arrangements, the semiconductor device **500** accordingly to the present embodiment may be configured as a dynamic random access memory (DRAM) device, and which may include at least one transistor (not shown in the drawings) and at least one capacitor (not shown in the drawings) to serve as the smallest memory cell of the DRAM array to receive the voltage signals from the second wires (namely, the bit lines) **560** and the first wires (namely, the word lines) **540**. The active areas **150** of the semiconductor device **500** also include the active fins **115** and the active ends **113** disposed at two sides of each active fin **115** and having a different material from that of the active fins **115**, so that, the plugs **590** may be stably disposed on the active fins **115**, active ends **113**, and the boundary therebetween. Accordingly, the plugs **590** may obtain a stable and reliable structure to achieve better electrically connection. In this way, the semiconductor device **500** of the present embodiment may therefore have improved structure and better functions.

(32) Please refer to FIGS. **11-12**, which illustrate schematic diagrams of a fabricating method of a semiconductor device according to a second embodiment in the present disclosure. The fabricating processes of the semiconductor device in the present embodiment is substantially the same as those in the aforementioned first embodiment, and all the similarities will not be redundantly described hereinafter. The difference between the fabricating method in the present embodiment and the aforementioned first embodiment is in that an insulating layer **320** is directly used as a mask layer in the present embodiment, to sequentially perform an etching process and a selectively epitaxial growing process.

(33) Precisely speaking, after defining a plurality of active area units **110** in the substrate **100**, the mask patterns (not shown in FIGS. **11-12**) are removed, and then, the insulating layer **320** is formed

to surround the active area units **110**. In other words, a top surface **320a** of the insulating layer **320** may be higher than the top surface **110a** of the active area units **110**, as shown in FIG. **11**.

(34) Next, a mask layer (not shown in the drawings) is formed on the substrate **100**, and which includes a plurality of openings (not shown in the drawings) in alignment with the gaps “g” between the adjacent ones of the active area units **110** respectively. Through the mask layer to perform the etching process, a plurality of openings **321** is formed in the insulating layer **320**, to expose the top surface **110a** and the sidewalls **110b** of the end portions **111** of each active area unit **110**, as shown in FIG. **12**. Following these, as being performed in FIGS. **5-7** in the aforementioned first embodiment, the selectively epitaxial growing process is performed while the insulating layer **320** being remained on the substrate **100**, so as to form the active ends **113** as shown in FIGS. **6-7**, or to form the active ends **117** as shown in FIG. **8**, but not limited thereto. Then, an planarization process (not shown in the drawings) is performed, to remove the insulating layer **320** disposed on the active area units **110**, so that, the insulating layer **320** may obtain a top surface **320b** lower than the top surface **110a** of the active area units **110**, to surround the active area units **110** to serve as the isolation structure. In this way, the active ends **113** or the active ends **117** may be formed in the present embodiment to improve the extending area of the active area, so as to enable the directly and stably connection between the storage node contacts and the active areas.

(35) Please refer to FIGS. **13-15**, which illustrate schematic diagrams of a fabricating method of a semiconductor device according to a third embodiment in the present disclosure. The fabricating processes of the semiconductor device in the present embodiment is substantially the same as those in the aforementioned first embodiment, and all the similarities will not be redundantly described hereinafter. The difference between the fabricating method in the present embodiment and the aforementioned first embodiment is in that a plurality of active area fragments **410** is formed through the SADP.

(36) Precisely speaking, as shown in FIGS. **13-14**, the active area fragments **410** are formed in the substrate **100** through the SADP or the SARP process, with each of the active area fragments **410** being parallel extended along the direction **D1** and being surrounded by a first insulating layer **420**. Next, a mask layer **430** is formed on the substrate **100**, and which includes a plurality of openings **431** to partially expose the active area fragments **410** underneath. Then, an etching process is performed through the mask layer **430**, to partially remove each of the active area fragments **410** which are exposed from each of the openings **431**, to form shallow trenches **102** in the substrate **100**. Accordingly, each of the active area fragments **410** may be cut into a plurality of active area units **110** as shown in FIG. **1**. In addition, after the etching process is performed, another etching process such a wet etching process may be performed then to partially remove the first insulating layer **420** disposed around the active area units **110**, especially the first insulating layer **420** closed to the end portions **111** of each active area units **110**, thereby forming the semiconductor device as shown in FIG. **3**

(37) After that, a selectively epitaxial growing process is performed while the mask layer **430** is remained on the substrate **100**, to form active ends **413** at two sides of each of the active area units **110**. Then, the rest portion of each of the active area units **110** form the active fins **415**, and the active ends **413** and the active fins **415** together form the active areas **450**, as shown in FIG. **15**. It is noted that, in the present embodiment, each of the active ends **413** may be formed on the side edges extending in the y-direction, and on the portion of the at least one side edge extending in the direction **D1** and being adjacent to the side edges extending in the y-direction, thereby presenting in an L-shape as being viewed from a top view (not shown in the drawings but being similar to the top view as shown in FIG. **6**). On the other hand, each of the active ends **413** is formed on the exposed surfaces at two sides of each active area units **110** (namely, including the sidewalls and the bottle walls of the shallow trenches **102** as shown in FIG. **15**), thereby presenting in an U-shape as being viewed from a cross-sectional view as shown in FIG. **15**. Then, after forming the active ends **413**, a second insulating layer **440** is formed in the shallow trenches **102**, and the second insulating layer

440 may include a top surface **440a** lower than the top surface **110a** of the active area units **110**, as shown in FIG. **15**. Otherwise, in another embodiment, the second insulating layer may optionally include a top surface being coplanar with the top surface **110a** of the active area units **110**. Accordingly, the remain portion of the first insulating layer **420** and the second insulating layer **440** may together form the isolation structure in the present embodiment. Through the aforementioned arrangements, the active ends **413** may also formed through the fabricating processes in the present embodiment, and which may improve the extending area of the active areas **450**, so as to enable the directly and stably connection between the storage node contacts and the active areas.

(38) Overall speaking, the semiconductor device of the present disclosure includes the active areas having composite materials. The active areas include the active fins and the active ends disposed at two sides of each active fin and having a different material from that of the active fins. The active ends is formed through the selectively epitaxial growing process, so that, each of the active areas may obtain an extended length thereby. In this way, the extending area of each active area is sufficiently enlarged, and the contact area between the active areas and the storage node contacts is also enlarged accordingly, so as to ensure the directly and stably connection between the active areas and the storage node contacts.

(39) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor device, comprising; a substrate; a plurality of active areas, separately with each other in the substrate, each of the active areas comprising an active fin and active ends disposed at two sides of the active fin, and the active fin and the active ends comprising different materials; an isolation structure, disposed in the substrate to surround the active areas; a plurality of first wires, disposed in the substrate to intersect with the active areas, across the isolation structure, wherein each of the first wires comprises: a gate electrode layer; and a capping layer disposed on the gate electrode layer; and a plurality of plugs disposed on the substrate, wherein at least one of the plugs directly contacts the active fin and one of the two active ends of one of the active areas, and the capping layer of one of the first wires.
2. The semiconductor device according to claim 1, wherein the active fin and the active ends comprise different roughness.
3. The semiconductor device according to claim 1, wherein the active ends comprise an epitaxial material.
4. The semiconductor device according to claim 1, wherein the active ends and the active fin have a height difference therebetween.
5. The semiconductor device according to claim 1, wherein the isolation structure comprises a first insulating layer and a second insulating layer, and the second insulating layer is disposed between adjacent ones of the active areas and is surrounded by the first insulating layer.
6. The semiconductor device according to claim 1, further comprising: a plurality of second wires, disposed on the substrate to intersect with the active areas, the second wires directly in contact with the active fins; a plurality of plugs disposed on the substrate, the plugs are alternately arranged with the second wires, wherein the plugs simultaneously contact the active fins and the active ends; and a spacer, disposed on the substrate, between each of the second wires and each of the plugs.
7. The semiconductor device according to claim 6, wherein bottom surfaces of the plugs disposed on the active fins and the active ends comprise a height difference.
8. The semiconductor device according to claim 6, further comprising: a plurality of contacts, disposed under the second wires, each of the contacts being disposed between adjacent ones of the

active ends.

9. A semiconductor device, comprising; a substrate; a plurality of active areas, separately with each other in the substrate, each of the active areas comprising an active fin and active ends disposed at two sides of the active fin, and the active fin and the active ends comprising different materials; an isolation structure, disposed in the substrate to surround the active areas; and a plurality of plugs, disposed on the substrate, at least one of the plugs directly in contact with the active fin and one of the two active ends of one of the active areas, a bottom plane of one of the plugs disposed on the active fin is lower than a bottom plane of the one of the plugs disposed on the one of the two active ends.
